

Problem Set 17

Problem 14.3.2:

(1) $\vec{\sigma} \cdot \hat{n} = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z$ with

$$(n_x, n_y, n_z) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

so

$$\vec{\sigma} \cdot \hat{n} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin \theta \cos \phi + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin \theta \sin \phi + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos \theta = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}.$$

So,

$$\vec{\sigma} \cdot \hat{n} |\hat{n}+\rangle = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix} = \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix} = |\hat{n}+\rangle,$$

where I used the trig identities:

$$\begin{aligned} \cos \theta \cos(\theta/2) + \sin \theta \sin(\theta/2) &= \cos(\theta - \frac{1}{2}\theta) = \cos(\theta/2), \\ \sin \theta \cos(\theta/2) - \cos \theta \sin(\theta/2) &= \sin(\theta - \frac{1}{2}\theta) = \sin(\theta/2). \end{aligned}$$

Similarly,

$$\vec{\sigma} \cdot \hat{n} |\hat{n}-\rangle = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} -\sin(\theta/2) e^{-i\phi/2} \\ \cos(\theta/2) e^{i\phi/2} \end{pmatrix} = \begin{pmatrix} \sin(\theta/2) e^{-i\phi/2} \\ -\cos(\theta/2) e^{i\phi/2} \end{pmatrix} = -|\hat{n}-\rangle.$$

(2) $\langle \hat{n} \pm | \vec{S} | \hat{n} \pm \rangle = \langle \hat{n} \pm | \frac{\hbar}{2} \vec{\sigma} | \hat{n} \pm \rangle = \langle \hat{n} \pm | \frac{\hbar}{2} (\sigma_x \hat{i} + \sigma_y \hat{j} + \sigma_z \hat{k}) | \hat{n} \pm \rangle$, so

$$\begin{aligned} \langle \hat{n}+ | \vec{S} | \hat{n}+ \rangle &= (\cos(\theta/2) e^{i\phi/2} \quad \sin(\theta/2) e^{-i\phi/2}) \frac{\hbar}{2} \begin{pmatrix} \hat{k} & \hat{i} - i\hat{j} \\ \hat{i} + i\hat{j} & -\hat{k} \end{pmatrix} \begin{pmatrix} \cos(\theta/2) e^{i\phi/2} \\ \sin(\theta/2) e^{-i\phi/2} \end{pmatrix} \\ &= \frac{\hbar}{2} \left[\hat{i} (\cos(\theta/2) \sin(\theta/2) (e^{i\phi} + e^{-i\phi})) + \hat{j} (-i \cos(\theta/2) \sin(\theta/2) (e^{i\phi} - e^{-i\phi})) \right. \\ &\quad \left. + \hat{k} (\cos^2(\theta/2) - \sin^2(\theta/2)) \right] = \frac{\hbar}{2} [\hat{i} \sin \theta \cos \phi + \hat{j} \sin \theta \sin \phi + \hat{k} \cos \theta], \end{aligned}$$

and similarly for $\langle \hat{n}- | \vec{S} | \hat{n}- \rangle$.

Problem 14.3.3: From

$$\sigma_i \sigma_j = -\sigma_j \sigma_i \quad (i \neq j) \tag{1}$$

$$\sigma_x \sigma_y = i \sigma_z \quad (\text{and cyclic}), \tag{2}$$

we have

$$\begin{aligned}
\text{Tr}(\sigma_z) &= \text{Tr}(-i\sigma_x\sigma_y) = -i\text{Tr}(\sigma_x\sigma_y) \\
&= -i\text{Tr}(-\sigma_y\sigma_x) = i\text{Tr}(\sigma_y\sigma_x) \\
&= i\text{Tr}(\sigma_x\sigma_y) \\
&= i\text{Tr}(i\sigma_z) = -\text{Tr}(\sigma_z),
\end{aligned}$$

and therefore $\text{Tr}(\sigma_z) = 0$, and similarly for σ_x and σ_y . In the above we used (1) to get the second line, (2) to get the first and fourth lines, and cyclicity of the trace to get the third line.

Problem 14.3.4:

(1) $\sigma_i\sigma_j = \frac{1}{2}[\sigma_i, \sigma_j]_+ + \frac{1}{2}[\sigma_i, \sigma_j] = \frac{1}{2}2\delta_{ij}I + \frac{1}{2}2i\sum_k \epsilon_{ijk}\sigma_k = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k$, therefore

$$\begin{aligned}
(\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma}) &= \sum_{ij} A_i B_j \sigma_i \sigma_j = \sum_{ij} A_i B_j \left(\delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k \right) \\
&= \left(\sum_i A_i B_i \right) I + i\sum_k \left(\sum_{ij} \epsilon_{ijk} A_i B_j \right) \sigma_k \\
&= (\vec{A}\cdot\vec{B})I + i\sum_k (\vec{A}\times\vec{B})_k \sigma_k = (\vec{A}\cdot\vec{B})I + i(\vec{A}\times\vec{B})\cdot\vec{\sigma}.
\end{aligned}$$

(2) Eqns. (14.3.42) and (14.3.43) imply that $M = \frac{1}{2}\sum_\alpha \text{Tr}(M\sigma_\alpha)\sigma_\alpha$ where $\sigma_\alpha = (\sigma_x, \sigma_y, \sigma_z, I)$. For $M = (\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma})$, we have $\text{Tr}(M\sigma_0) = \text{Tr}(MI) = \text{Tr}(M) = \sum_{ij} A_i B_j \text{tr}(\sigma_i\sigma_j) = 2\sum_{ij} A_i B_j \delta_{ij} = 2\vec{A}\cdot\vec{B}$, and

$$\begin{aligned}
\text{Tr}(M\sigma_k) &= \text{Tr}((\vec{A}\cdot\vec{\sigma})(\vec{B}\cdot\vec{\sigma})\sigma_k) = \sum_{ij} A_i B_j \text{Tr}(\sigma_i\sigma_j\sigma_k) \\
&= \sum_{ij} \frac{1}{2} A_i B_j \text{Tr}(\sigma_k\sigma_i\sigma_j + \sigma_i\sigma_j\sigma_k) \quad (\text{using cyclicity of the trace}) \\
&= \sum_{ij} \frac{1}{2} A_i B_j \text{Tr}([\sigma_k, \sigma_i]\sigma_j + \sigma_i[\sigma_k, \sigma_j]_+) \\
&= \sum_{ij} \frac{1}{2} A_i B_j \text{Tr}(2i\sum_\ell \epsilon_{kil}\sigma_\ell\sigma_j + 2\sigma_i\delta_{kj}) \\
&= i\sum_{ij\ell} A_i B_j \epsilon_{kil} \text{Tr}(\sigma_\ell\sigma_j) \quad (\text{since } \text{Tr}(\sigma_i) = 0) \\
&= i\sum_{ij\ell} A_i B_j \epsilon_{kil} 2\delta_{\ell j} = 2i\sum_{ij\ell} A_i B_j \epsilon_{ijk} \quad (\text{using } \epsilon_{ijk} = \epsilon_{kij}) \\
&= 2i(\vec{A}\times\vec{B})_k.
\end{aligned}$$

So

$$\begin{aligned}
 (\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma}) &= \frac{1}{2} \sum_{\alpha} \text{Tr}(M \sigma_{\alpha}) \sigma_{\alpha} = \frac{1}{2} \text{Tr}(M \sigma_0) I + \frac{1}{2} \sum_k \text{Tr}(M \sigma_k) \sigma_k \\
 &= \frac{1}{2} (2 \vec{A} \cdot \vec{B}) I + \frac{1}{2} \sum_k 2i (\vec{A} \times \vec{B})_k \sigma_k = (\vec{A} \cdot \vec{B}) I + i (\vec{A} \times \vec{B}) \cdot \vec{\sigma}.
 \end{aligned}$$

Problem 14.3.5:

$$\begin{aligned}
 M &= \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \frac{\alpha+\delta}{2} + \frac{\alpha-\delta}{2} & \frac{\beta+\gamma}{2} - i \left(i \frac{\beta-\gamma}{2} \right) \\ \frac{\beta+\gamma}{2} + i \left(i \frac{\beta-\gamma}{2} \right) & \frac{\alpha+\delta}{2} - \frac{\alpha-\delta}{2} \end{pmatrix} \\
 &= \left(\frac{\alpha+\delta}{2} \right) I + \left(\frac{\beta+\gamma}{2} \right) \sigma_x + i \left(\frac{\beta-\gamma}{2} \right) \sigma_y + \left(\frac{\alpha-\delta}{2} \right) \sigma_z.
 \end{aligned}$$